



Basic reason for the accumulation of charge on the surface of polymer dielectrics

Tianyu Wang^{1†}, Xiaofen Li^{2†}, Boya Zhang³, Dayu Li¹, Jianbo Liu^{2*} and Guixin Zhang^{1*}

Due to their outstanding electrical insulation performance, low energy loss and high reliability, polymer dielectrics are widely used in flexible electronics, capacitors and large-scale power transmission equipment [1–5]. However, polymer dielectrics accumulate large amounts of charge on their surface during use, which causes local electric field distortion and induces flashover, explosion, and equipment damage [6–9]. Although researchers have attempted to develop methods to reduce the accumulation of surface charge through material modification, such as plasma treatment [10], direct fluorination [11], and coating [7,9,12], each of these methods has drawbacks; thus, the bottleneck has yet to be overcome. The reason for this challenge is that it remains unclear what the fundamental cause of charge accumulation is, that is, what the composition of the polymer charge carrier traps is.

The charge characteristics of polymer dielectrics are directly related to their trap characteristics [9,13–15]. During the preparation of polymer dielectrics, a localized state with an energy level in the forbidden band is formed inside the polymer dielectric; that is, carrier traps are formed [16–18]. These traps represent sites in the polymer that allow carriers to stay, which provides conditions for charge accumulation. The traps can be divided into deep traps and shallow traps according to their energy levels. Carriers in deep traps are easy to capture but difficult to remove. In contrast, carriers in shallow traps easily escape but are difficult to capture. Therefore, it can be considered that the deep traps in the polymer dielectric lead to the accumulation of surface charges, and the large accumulation of surface charges will lead to the reduction of the flashover voltage. The deep and shallow traps in polymers are caused by chemical and physical defects, respectively [16–18]. Therefore, in theory, the most ideal way to control the charge is to first study the specific groups or chemical components caused by the chemical defects in the polymer and to eliminate these groups or chemical components through certain reagents during the preparation process. In so doing, a reduction in the number of deep traps and a decrease in surface charge accumulation could be achieved.

In this letter, we describe a method we have developed to study the microscopic charge characteristics of polymers for studying the components of deep traps. We use magnetic force microscopy (MFM) to observe the distribution of radicals on the

polymer dielectric surface and Kelvin probe force microscopy (KPFM) to observe the charge characteristics of radicals in the polymer on a subnanometer scale. It is found that the radicals exhibit the characteristics of deep traps. Then, through first-principles calculations, it is found that there is a defect state in the band gap of the polymer after the radicals appear. This defect state is in the depth of the band gap, indicating that there is a deep level defect state in the system at this time. Furthermore, after treating the polymer with a radical scavenger, we find through microscopic and macroscopic charge measurement methods that the amount of charge accumulated on the surface of the polymer has been significantly reduced. This finding has important theoretical and practical significance, and the test and analysis methods of the microcharge characteristics and distribution in this study have important reference value for researchers in related fields.

Because of its good insulation, chemical stability and mechanical properties, epoxy resin (EP) is widely used in the electronics industry and in large-scale electrical equipment, so we chose EP as the main research object in this study. First, the distribution of radicals was observed through MFM [6,19–21]; the result is shown in Fig. 1a. The light-colored area in the figure is the radical area [6,19–21] (see Figs S1 and S2 in the Supplementary information for the other scan scales). KPFM was used to test the charge dissipation characteristics of the radical region and the nonradical region; the results are shown in Fig. 1b–e. As shown in Fig. 1b, c, the charge in the radical region dissipates very little except at the precise moment the bias voltage is removed. The rapid drop in potential just after the bias voltage is removed may be caused by the rapid dissipation of charges that are freed between the material and probe and not captured by free radicals. As shown in Fig. 1d, e, when the bias voltage is removed, the charge in the nonradical region completely dissipates in approximately 140 s. The above results indicate that radicals easily capture and stabilize charges. Fig. S2 shows the distribution of charge and radicals at the same position. Notably, their distributions are basically the same, which explains the role of radicals in capturing and stabilizing the charge. Based on Fig. 1c, e, the dissipation curve of the potential after the bias voltage is removed was combined with the isothermal surface potential decay (ISPD) method [7,17,18], which allowed the trap parameters in the radical and nonradical regions to be obtained.

¹ State Key Laboratory of Control and Simulation of Power System and Generation Equipment, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China

² Key Laboratory of Advanced Materials (MOE), School of Materials Science and Engineering, Tsinghua University, Beijing 100084, China

³ State Key Laboratory of Electrical Insulation and Power Equipment, Xi'an Jiaotong University, Shaanxi 710049, China

[†] These authors contributed equally to this work.

* Corresponding authors (emails: jbliu@mail.tsinghua.edu.cn (Liu J); guixin@mail.tsinghua.edu.cn (Zhang G))

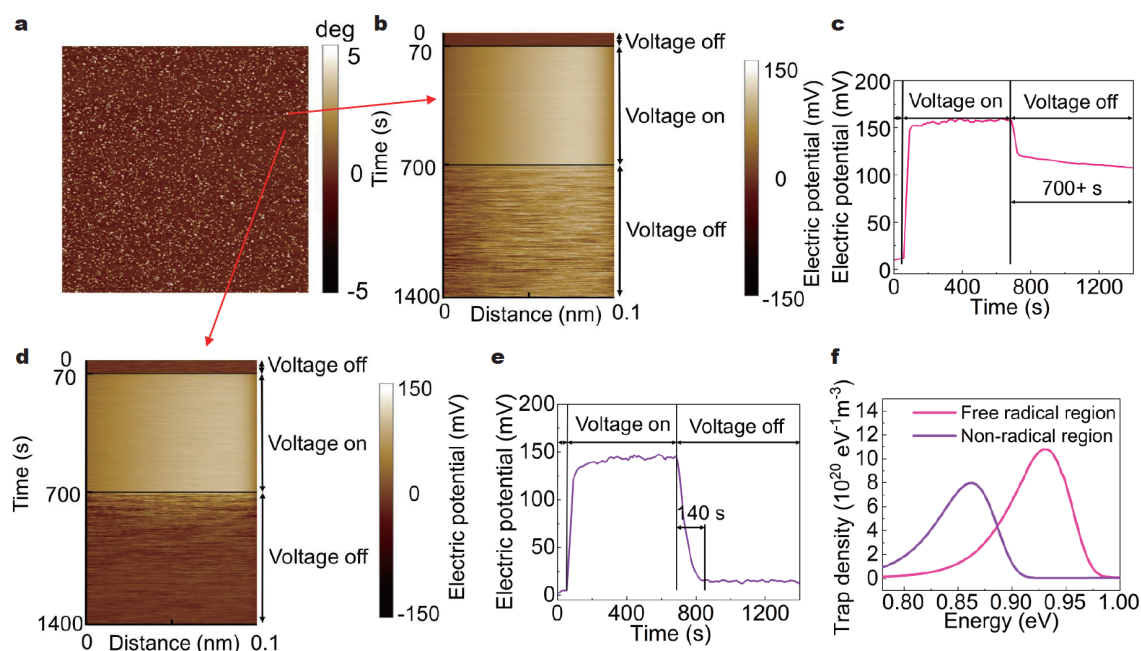


Figure 1 Charge characteristics and trap parameters of the radicals in EP. (a) MFM map of the EP surface (scanning area, $5\ \mu\text{m} \times 5\ \mu\text{m}$). (b) KPFM map and (c) value of the potential in the radical region. (d) KPFM map and (e) value of the potential in the non-radical region. (f) Deep trap parameters in the radical and non-radical regions.

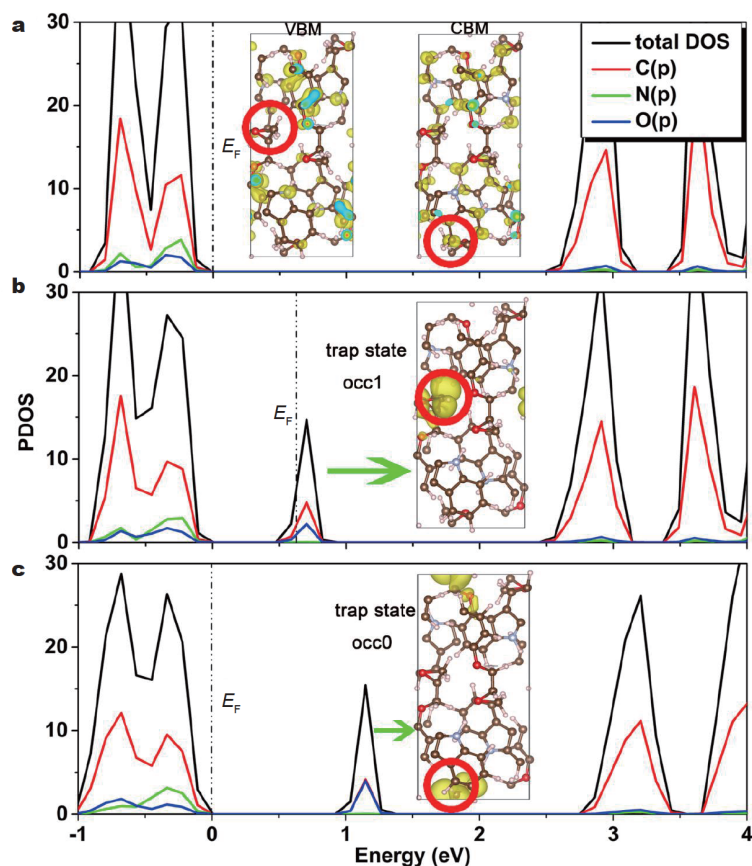


Figure 2 First-principles calculation results. PDOS of (a) pure EP, (b) EP with the removal of H, and (c) EP with the removal of CH_2 . occ1 is a trap state occupied by one electron, occ0 is a trap state occupied by zero electron. The insets show the partial charge densities of band edges of pure EP and the trap states occ1 and occ0 of EP with the removal of H and CH_2 , respectively. The positions before and after removing H and CH_2 are indicated by the red circles. Yellow isosurfaces denote areas of charge accumulation and depletion ($\pm 0.003\ \text{e}\ \text{\AA}^{-3}$). The valence band maximum (VBM) is set to zero. Color code: C, brown; H, pink; N, gray; O, red.

The results are shown in Fig. 1f. The trap energy level of the nonradical region is approximately 0.86 eV, while the trap energy level of the radical region is approximately 0.94 eV. These results directly show that the radicals in EP exhibit deep trap behaviors.

The radicals of polymer dielectrics, such as EP, are mainly formed by the random cleavage of weak bonds, such as C–H [22]. Therefore, our calculations are for EP after radicals have been generated by the removal of H and CH₂ in the EP molecule. We observe that neither the deep defect state nor the edge zone appear in Fig. 2a, which means that electron hole recombination is less likely. However, the defect state is occupied by a single electron (named occ1) and is occupied by no electron (named occ0) of the EP after radicals are generated in Fig. 2b, c, respectively. We find that the defect states are all deep within the bandgap. The corresponding partial charge densities (Fig. 2b, c) of trap states occ1 and occ0 become localized, which match the projected density of states (PDOS). These results show that there are deep-level defect states in the material system for this case.

To verify the conclusion that the radicals are polymer charge carrier deep traps, we performed macro- and microscale charge distribution measurement experiments. Fig. 3a–d show the

results of the microcharge experiments performed with scanning electron microscopy (SEM). Fig. 3a, b show the pristine EP results. Fig. 3c, d show the results of EP treated with radical scavengers. The radical scavenger was *n*-octadecyl- β -(4-hydroxy-3,5-di-*tert*-butyl-phenyl)-propionate. For specific processing methods, see the Supplementary information, specifically the Materials and Preparation sections. For pristine EP, the SEM electrons are captured after hitting the surface and deposited into a charge spot with a high electron density. However, the electron distribution of the treated EP is uniform, and the electron density is greatly reduced.

To examine the charge distribution at the macroscale, we used the needle-plate electrode corona discharge method to inject positive charges on the surface of the pristine and treated EP samples. The charge distribution is shown in Fig. 3e, f. The amount of charge accumulated on the surface of EP after radical removal is greatly reduced, and the distribution is more uniform, which is consistent with the results of the SEM charge experiment. The numerical distribution of the charge density is shown on the black dashed line in Fig. 3g, and it can be seen that the highest charge density drops from 66.32 to 8.12 pC mm⁻² after treatment. The experimental results of the negative charge

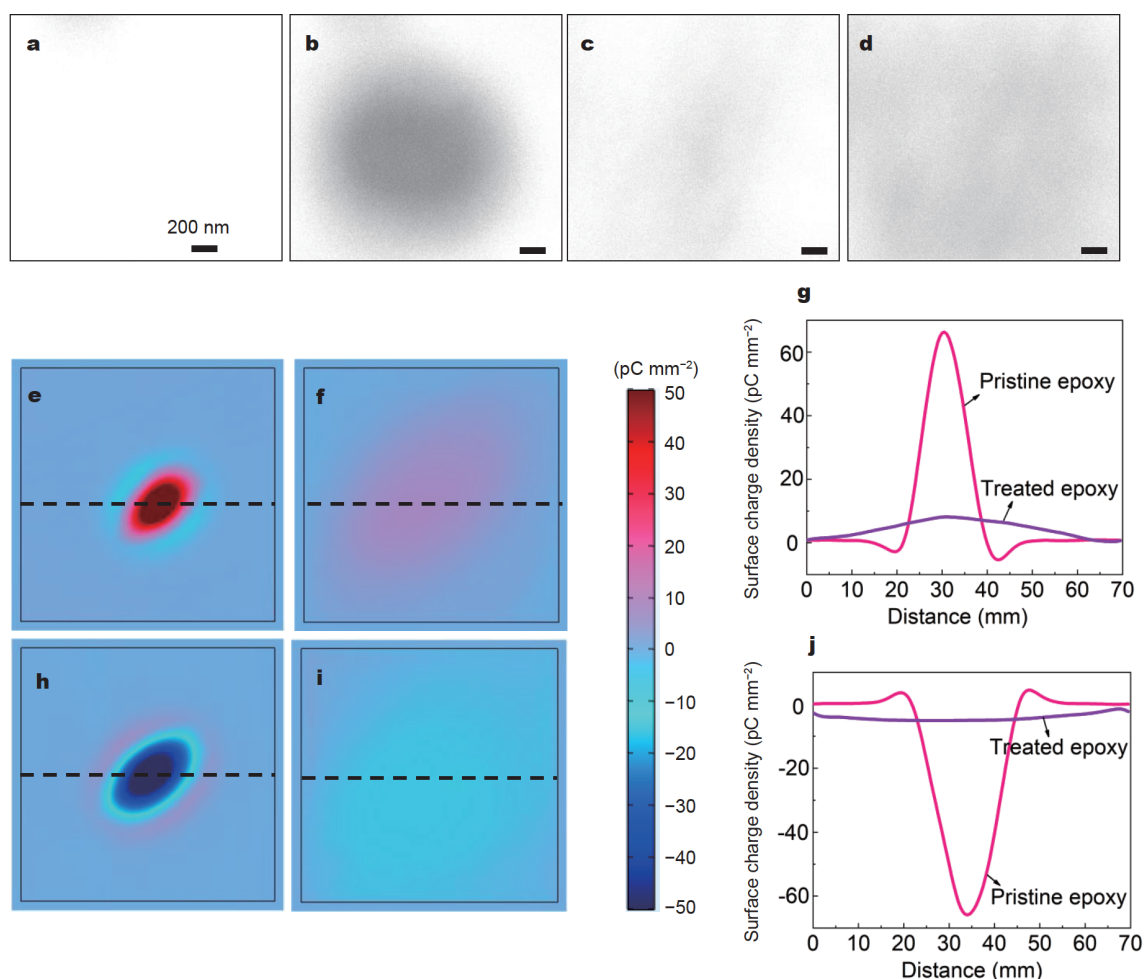


Figure 3 Micro- and macrocharge characterization. Pristine EP was charged with SEM for (a) 0 and (b) 60 s, and EP treated with radical scavenger was charged with SEM for (c) 0 and (d) 60 s. The surface charge density distribution map of (e) pristine epoxy and (f) treated EP charged with needle-plate electrode corona discharge and the application of a positive voltage (scanning area, 70 mm × 70 mm). (g) Numerical distribution of the charge density on the black dashed line in (e, f). The surface charge density distribution map of (h) pristine epoxy and (i) treated epoxy charged with needle-plate electrode corona discharge and the application of a negative voltage. (j) Numerical distribution of the charge density on the black dashed line in (h, i).

injection are shown in Fig. 3h–j. Similar to the pattern observed for positive charge, the highest charge density changes from -65.86 to -5.04 pC mm $^{-2}$. The above experimental results provide further verification that radicals are polymer charge carrier deep traps. Since the amount of surface charge accumulated on EP is significantly reduced after removing the radicals, the flashover voltage is significantly increased (Fig. S3). In particular, when the polymer surface is precharged, the flashover voltage is increased by approximately 72.8%. This increase is much greater than that achieved by the various material modification schemes to increase flashover voltage reported in the literature [7].

In addition, we compared the charge density and flashover voltage before and after scavenging radicals of a biaxially oriented polypropylene (BOPP) film, which is currently widely used in energy storage polymer dielectrics, and of a polyimide (PI) film, which is widely used in many engineering fields (Figs S4–S10). The charge density is significantly reduced after scavenging the radicals, indicating that the function of radicals in polymers as deep traps has some degree of universality.

In summary, this letter describes the charge characteristics of radicals in the nano- and microdomains of EP observed based on KPFM and MFM and reveals that radicals exhibit deep trap characteristics. Through first-principles calculations, we found that there are deep-level defect states in the system after the emergence of radicals, which may explain the above experimental findings. Subsequently, EP was treated with a radical scavenger. Through microscopic and macroscopic charge measurement methods, we found that the amount of charge accumulated on the polymer surface significantly decreases while the flashover voltage significantly increases. This treatment also achieved good results with BOPP and PI. This article describes the first use of advanced experimental methods combined with theoretical calculations to reveal that radicals in polymer dielectrics are deep traps. This research can guide subsequent studies on the charge characteristics of polymer dielectrics. Additionally, this research demonstrates that we can effectively reduce surface charge accumulation by using free radical scavengers and thus provides a reference for solving related engineering problems.

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Author contributions Wang T conceived the idea, Wang T and Zhang G designed the experiments, Wang T carried out the experiments, Li X and Liu J carried out the DFT calculations, and Wang T, Li X, Zhang B, and Li D analyzed the data. Wang T and Li X wrote the manuscript. All authors discussed the results and commented on the manuscript.

Conflict of interest The authors declare that they have no conflict of interest.

Supplementary information Experimental details and supporting data are available in the online version of the paper.



Tianyu Wang was born in Gansu province, China in 1995. He is now working towards his PhD degree at the Department of Electrical Engineering, Tsinghua University. His research interests focus on nanometric dielectrics and the surface charge accumulation of polymer materials.



Xiaofen Li received a bachelor's degree in materials science and engineering from Nanchang University in 2019. She is now a PhD candidate at the Department of Materials Science and Engineering, Tsinghua University. Her current research interests focus on using the first-principle calculations to study the defect properties and ferroelectric polarization properties of perovskite photoelectric materials.



Jianbo Liu is an associate professor at the School of Materials Science and Engineering, Tsinghua University. He earned his doctoral degree from the Department of Materials Science and Engineering at Tsinghua University in 2002. He did postdoctoral research at the Department of Materials Science and Engineering, University of Illinois at Urbana-Champaign from 2003 to 2008. His research interests include simulation of clean energy storage materials, such as perovskite photovoltaic materials and lithium-ion battery electrode materials.



Guixin Zhang was born in Tianjin, China in 1963. He received his BS and MS degrees from the Department of Electrical Engineering, Tsinghua University in 1987 and 1993, respectively, and a PhD degree from the Department of Physics, Nanyang Technological University in Singapore in 1999. His current research interests are polymer dielectrics, plasma, high voltage insulation, optical fiber sensor, and electronic current/voltage transformer.

聚合物电介质表面电荷积聚的根本原因

王天宇^{1†}, 李小芬^{2†}, 张博雅³, 李大雨¹, 刘剑波^{2*}, 张贵新^{1*}

摘要 聚合物电介质在使用过程中表面会积聚大量的电荷, 导致局部电场畸变、闪络、爆炸和设备损坏等问题。由于造成电荷积聚的聚合物载流子深陷阱的主要构成目前仍不清楚, 这一瓶颈问题目前仍未得到有效解决。本文基于开尔文探针力显微镜(KPFM)和磁力显微镜(MFM)观测了纳米微区聚合物自由基的电荷行为, 发现自由基表现出深陷阱的特性。第一性原理计算发现出现自由基后的体系中存在深能级缺陷态。微观和宏观的电荷测量表明, 清除自由基后的聚合物表面电荷量显著减少, 这为聚合物电介质在诸多领域中的安全使用提供了有效的材料改性方案。本文首次揭示了电介质中的自由基是深陷阱, 为后续关于聚合物电介质的电荷特性方面的研究提供了重要的理论指导。